

BhoomiSetu: The Digital Bridge for National Land Acquisition & Management

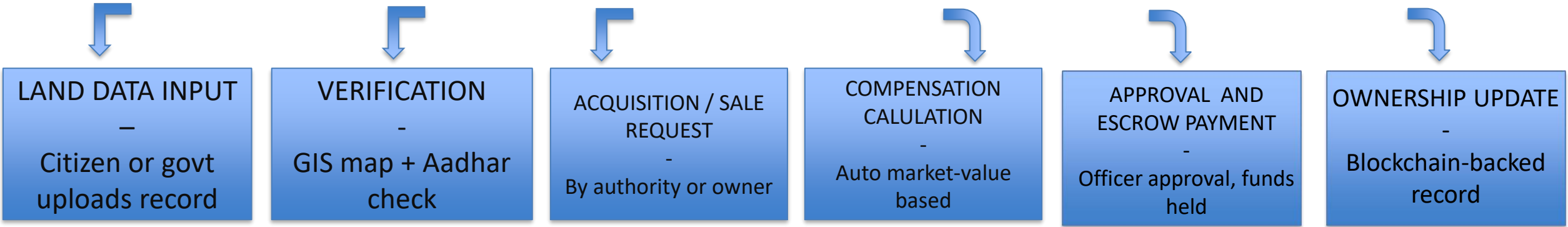
- Problem Statement ID – SIH26016
- Problem Statement Title -
Real-Time National Land Acquisition &
Management System for End-to-End Digital
Monitoring and Decision Support
- Theme- Smart Automation
- PS Category - Software
- Team ID- SIH26016
- Team Name - Syntax



PROPOSED SOLUTION-
REAL-TIME NATIONAL LAND ACQUISITION AND
MANAGEMENT SYSTEM

BhoomiSetu

A unified digital platform for real-time land ownership, acquisition, and settlement
– connecting citizens, government, and financial institutions

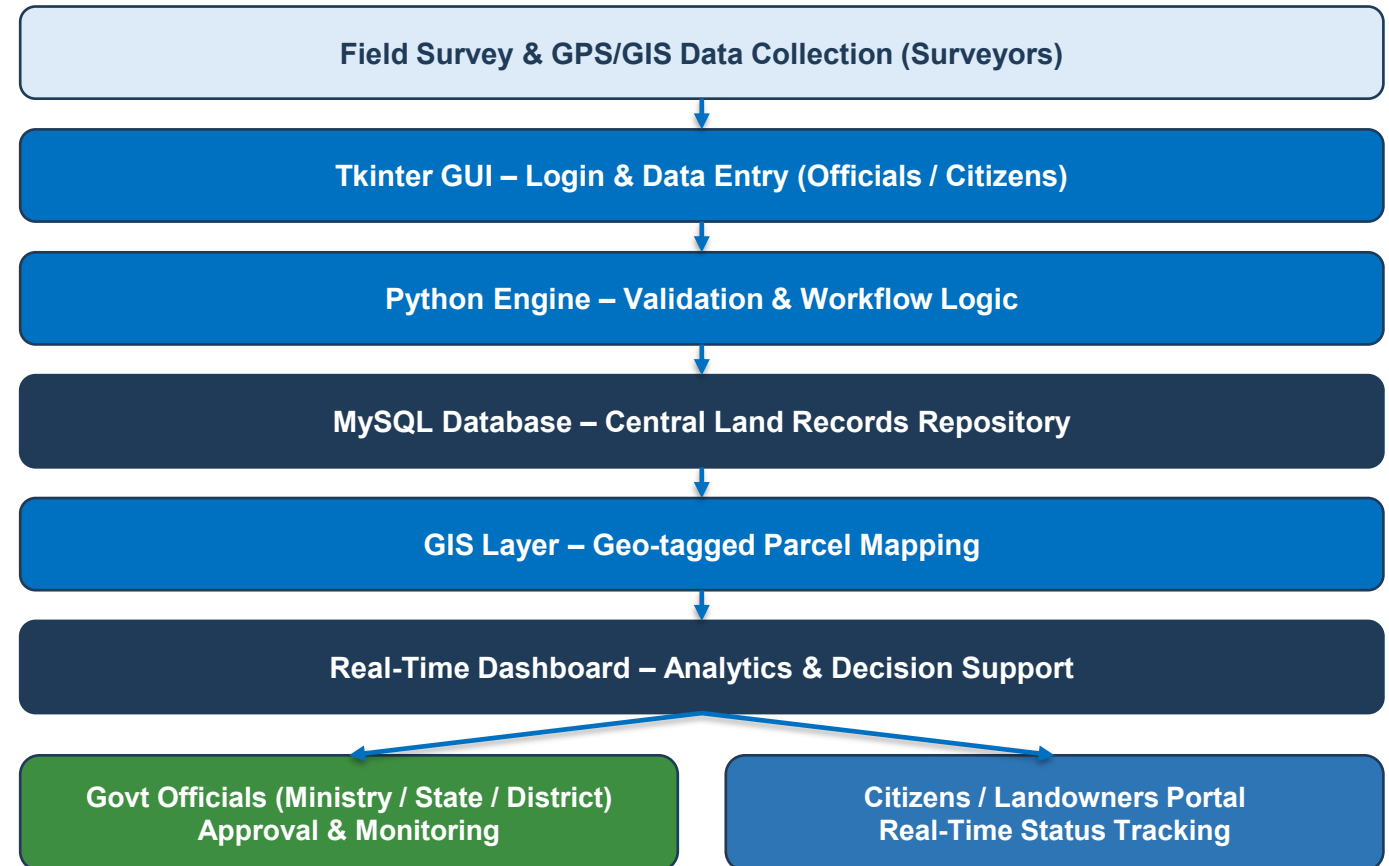


PROBLEM AT HAND	HOW WE MAKE A DIFFERENCE
Records scattered across departments	Single unified platform
Compensation takes months	Automated, instant calculation
No visibility into the case status	Real-time online tracking
Manual, error- prone process	Digital, tamper-proof

- **TECHNOLOGY STACK & WHY IT'S EFFICIENT**

- **Python** (Application Logic) – Rapid development, huge ecosystem (Pandas, GeoPandas), easy MySQL & GIS integration, fully cross-platform.
- **MySQL** (Database Backend) – ACID-compliant & reliable, fast indexed queries on large land-record datasets, proven scalability, free & open-source.
- **Tkinter** (Desktop GUI) – Built into Python (zero extra setup), lightweight & responsive, ideal for offline-capable use by field & office staff.
- **GIS Layer – GeoPandas/QGIS** (Spatial Mapping) – Enables geo-tagged parcel visualization & boundary verification for accurate land monitoring.

SYSTEM ARCHITECTURE & PROCESS FLOW



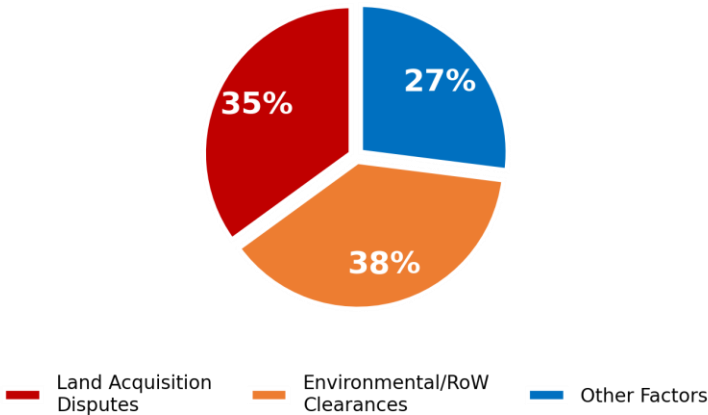
Technical Feasibility	Economic Feasibility	Operational Feasibility
Built on mature infra: Bhuvan/ISRO, DILRMP, Bhu-Naksha, Bhu-Aadhaar, Aadhaar verification.	Cuts recurring cost of manual surveys, paperwork & litigation from acquisition delays.	Needs coordinated onboarding of Revenue, Registration, Survey & local bodies onto one platform.
Challenge: unifying fragmented, paper-based, regional-language state records into one layer.	Upfront cost far lower than current cost of stalled projects (₹1.92 lakh cr, 2016 study).	Compliant by design — mirrors RFCTLARR Act, 2013 (notify → survey → award → compensate).
Real-time GIS/satellite tracking gives live status from survey to possession across states.	Cost recovery potential through reduced litigation spend and faster project execution.	Built-in grievance redressal & audit trails ensure transparency and accountability at each stage.



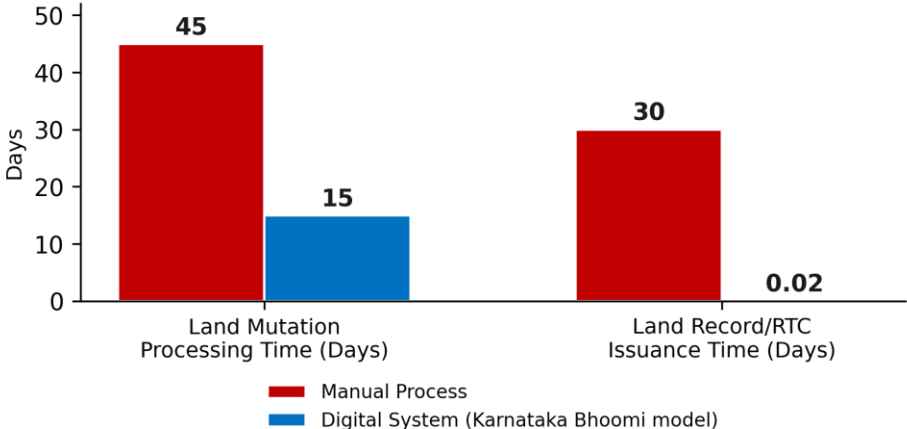
GOOD TO KNOW

Nearly two-thirds of all civil cases in India involve land or property disputes, and a land acquisition case reaching the Supreme Court takes ~20 years on average to resolve — a delay this system is built to prevent through real-time, auditable tracking.

Causes of Infra Project Delays in India



Manual vs Digital: Process Time (Days)



Sources: PRAGATI / Rau's IAS 2026 (delay causes); Dept. of Land Resources, DILRMP MIS 2025 (digitization); Karnataka Bhoomi portal — RTC & auto-mutation timelines (2026); PRS India, IIPER (dispute resolution time)

“From a fragmented, paperwork-driven process to a transparent, real-time and data-driven land acquisition ecosystem.”



System Transformation

Metric	Legacy Manual Process	Real-Time Digital System
Data Sync	Weeks of postal and ledger entry delays	Instant sync via secure GIS and cloud APIs
Fraud Risk	High risk of duplicate or altered records	Immutable audit logs preventing unauthorized edits
Payout Speed	Months of routing through bureaucratic silos	Direct Benefit Transfer on milestone approval
Disputes	Lengthy court cases over vague boundaries	AI-driven overlap detection using geo-fencing
Reporting	Static, outdated spreadsheet summaries	Dynamic analytics with live bottleneck tracking
Project Tracking	Fragmented offline records with limited visibility across departments	End-to-end visibility across every project from proposal to completion in one platform
Decision-Making	Slow administrative movement leading to unnoticed bottlenecks	Faster decision-making via real-time dashboards and early alerts for delays
Grievance Resolution	Paper complaints requiring manual follow-ups and long turnaround times	Faster resolution via real-time digital tracking and automated escalation

Impact & Benefits



Social Benefits

- Empowers Local Landowners
- Secures Direct Payouts
- Prevents Boundary Litigation



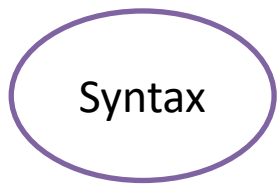
Economic Benefits

- Stops Budget Overruns
- Accelerates Infrastructure Growth
- Cuts Administrative Costs



Environmental Benefits

- Protects Ecological Reserves
- Optimizes Land Allocation
- Eliminates Paper Waste



RESEARCH AND REFERENCES



Details / Links of the reference and research work

Slide / Claim Referenced	Source	Link
Technical Approach — GIS layer (Bhuvan/ISRO)	Bhuvan, ISRO's National Geoportal (NRSC)	http://bhuvan.nrsc.gov.in/
Feasibility — DILRMP, Bhu-Naksha, Bhu-Aadhaar (ULPIN) infrastructure	Dept. of Land Resources — DILRMP Programme Page	https://dolr.gov.in/en/programmes-schemes/dilrmp-2/
Feasibility — RFCTLARR Act, 2013 process (notify → survey → award → compensate)	Dept. of Land Resources — Acts & Rules	https://dolr.gov.in/en/act-rules/
Feasibility — RFCTLARR Act, 2013 compensation provisions	PRS Legislative Research — Bill Summary	https://prsindia.org/billtrack/the-right-to-fair-compensation-and-transparency-in-land-acquisition-rehabilitation-and-resettlement-bill-2013
Feasibility — "Nearly two-thirds of civil cases involve land/property disputes"	DAKSH — Access to Justice Survey, 2016	https://dakshindia.org/Justice-Frustrated/litigation-in-land-acquisition
Feasibility — Manual vs. digital land-record processing times	Government of Karnataka — Bhoomi Portal (RTC & Auto-Mutation)	https://www.thelandbankers.com/bhoomi-karnataka
Feasibility — Land acquisition as leading cause of infra project delays	PRAGATI-linked analysis, Rau's IAS Compass	https://compass.rauias.com/current-affairs/breaking-ground-land-acquisition-slows-india-infrastructure-push/